

1. A circuit element resists a sudden change, but not the one most beginners first assume.

In steady DC it becomes almost invisible, yet in switching circuits it dominates behavior.

What is it?

Ans: Inductor

2. You are given a device that never stores energy, only dissipates it, yet it is essential to protect almost every other component.

Remove it from an LED circuit, and failure becomes immediate.

Name this silent limiter.

Ans: Resistor

3. A device shows zero voltage across it at an instant, yet it is not necessarily inactive.

Another shows zero current but may still not be “off.”

What physical quantity resolves this contradiction?

Ans: Power

4. In a communication system, a signal goes on a long journey and “gets tired,” not because of distortion but due to gradual weakening.

What is this reduction in strength formally called?

Ans: Attenuation

5. A digital technology where ideally, no current flows except during switching—making it extremely power efficient.

Yet, it dominates modern processors and IC design.

What is this technology?

Ans: CMOS

6. A system produces a periodic output without any external periodic input.

It neither amplifies a constant signal nor remains stable.

Instead, it feeds on its own feedback.

What phenomenon is this?

Ans: Oscillation

7. Two systems carry signals. One is louder, the other carries more “information per second.”

The second system is said to have more of this quantity.

What is it?

Ans: Bandwidth

8. A material that behaves like neither a perfect conductor nor a perfect insulator, yet forms the foundation of modern electronics.

Its behavior can be altered dramatically by adding tiny impurities.

Identify it.

Ans: Semiconductor

9. Two sine waves look identical in frequency and phase, but one appears “taller” on the graph.

What property distinguishes them?

Ans: Amplitude

10. You apply electrical energy, and instead of heat or light, you get continuous mechanical rotation.

What device performs this conversion?

Ans: Motor

11. In AC circuits, resistance alone fails to explain opposition to current.

A more complete quantity combines resistance and reactance into a single term.

Name it.

Ans: Impedance

12. I must be complete for anything to happen.

Break me anywhere, and everything stops instantly.

Yet I am not a single component.

What am I?

Ans: Circuit

13. You feed in an AC signal expecting it to alternate, but the output refuses to go negative.

Which process (or device function) is responsible?

Ans: Rectifier

14. In a transistor, increasing input no longer increases output beyond a point.

The device is fully “on,” yet no further control is possible.

What region is this called?

Ans: Saturation

15. Electrons do not move without a reason.

This quantity provides the “push” that drives them through a conductor.

What is it?

Ans: Voltage

16. A signal that can take infinitely many values within a range, unlike its discrete counterpart used in digital systems. What type of signal is this?

Ans: Analog

17. A high-frequency signal is sampled too slowly and suddenly appears as a completely different lower frequency. What deceptive phenomenon is occurring?

Ans: Aliasing

18. A three-terminal device that can act as a switch or an amplifier, forming the backbone of modern electronics. Without it, digital computing as we know it would not exist. Identify it.

Ans: Transistor

19. A component that behaves like a one-way valve for current—allowing flow in one direction and blocking the other. What is it?

Ans: Diode

20. The relationship between voltage and current was formalized into a law named after a scientist. What is the unit (and name) associated with this law?

Ans: Ohm











